

NEURUPPIN, Germany

WMO: 102700

Lat: 52.9037N

Lon: 12.8071E

Elev: 39

StdP: 100.86

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-13.1	-10.3	-15.9	0.9	-12.1	-13.3	1.2	-9.6	11.1	1.9	10.1	0.8	2.2	70	0.558

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.6	28.4	19.7	26.7	18.7	25.1	18.1	20.6	26.6	19.6	25.0	18.7	23.6	3.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.5	13.4	23.1	17.5	12.6	22.1	16.6	11.9	21.4	59.3	26.8	56.1	25.1	53.0	23.7	24.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.5	8.3	7.3	-15.0	32.1	5.4	1.9	-18.9	33.5	-22.1	34.6	-25.1	35.7	-29.1	37.1		
(5)			-15.3	23.3	5.3	0.8	-19.2	23.9	-22.3	24.4	-25.3	24.9	-29.2	25.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.0	0.4	-0.5	3.7	7.8	13.0	15.9	18.1	17.8	14.1	9.7	4.6	2.0
	DBStd	7.50	5.40	4.80	3.54	3.53	3.62	3.36	3.13	3.06	2.61	3.58	3.30	3.71
	HDD10.0	1343	296	296	194	84	13	0	0	0	1	48	163	248
	HDD18.3	3514	554	529	452	314	169	89	44	47	129	267	412	506
	CDD10.0	962	0	0	1	19	106	176	251	243	124	40	1	0
	CDD18.3	90	0	0	0	0	3	16	37	32	3	0	0	0
	CDH23.3	766	0	0	0	1	30	145	323	247	21	1	0	0
	CDH26.7	151	0	0	0	0	1	27	68	54	1	0	0	0
Wind	WSAvg	3.3	4.1	3.6	3.7	3.3	3.1	2.9	2.8	2.6	2.9	3.2	3.7	3.9
Precipitation	PrecAvg	602	51	40	40	31	54	63	73	60	50	45	44	51
	PrecMax	870	116	97	84	65	160	134	234	138	146	121	88	93
	PrecMin	413	1	8	9	1	13	17	26	23	12	9	3	11
	PrecStd	101	28	19	21	15	33	29	45	29	27	26	21	19
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.4	10.9	14.9	21.5	25.7	29.6	30.6	31.0	25.5	21.2	13.0	11.3
		MCWB	9.0	8.5	10.4	13.0	17.3	19.6	21.0	21.5	18.6	16.4	10.9	9.3
	2%	DB	8.9	8.2	12.1	18.5	24.1	27.1	28.6	27.8	23.3	18.0	11.7	10.0
		MCWB	7.3	6.3	8.3	12.0	16.4	18.2	19.8	19.9	17.5	14.3	10.1	8.7
	5%	DB	7.8	6.4	10.1	16.1	22.2	24.8	26.8	26.1	20.7	16.2	10.3	8.7
		MCWB	6.3	4.9	7.3	10.8	15.1	17.2	18.9	19.0	15.9	13.2	8.9	7.5
	10%	DB	6.5	5.2	8.6	14.0	20.2	22.7	25.1	24.1	19.0	14.8	9.2	7.5
		MCWB	5.2	3.8	6.3	9.6	14.1	16.2	18.4	18.0	15.0	12.2	8.0	6.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.1	9.3	10.8	14.0	18.5	21.2	22.5	22.3	19.5	17.1	11.2	9.8
		MCDB	10.8	10.8	13.8	19.2	23.7	26.8	29.2	29.5	24.2	20.2	12.3	10.7
	2%	WB	7.8	6.7	9.0	12.3	17.1	19.4	20.6	20.5	18.1	15.1	10.2	8.7
		MCDB	8.7	8.0	11.3	17.6	23.3	24.7	26.7	26.7	21.8	17.4	11.5	9.8
	5%	WB	6.3	5.1	7.6	11.2	15.8	18.1	19.6	19.5	16.7	13.7	9.1	7.6
		MCDB	7.5	6.2	9.6	15.6	21.1	23.4	25.5	24.6	19.9	15.8	10.0	8.6
	10%	WB	5.3	3.9	6.5	9.9	14.4	16.8	18.5	18.6	15.6	12.5	8.1	6.3
		MCDB	6.5	4.9	8.5	13.8	19.1	21.7	23.8	23.1	18.4	14.3	9.1	7.3
Mean Daily Temperature Range	5% DB	MDBR	4.5	5.9	6.5	8.4	9.4	9.1	9.6	9.4	8.2	6.9	4.6	4.1
		MCDBR	4.6	6.5	9.0	11.6	12.2	12.6	12.8	12.3	11.0	8.6	5.6	5.1
		MCWBR	4.0	5.2	6.2	6.5	6.6	6.2	6.2	6.3	6.4	5.7	4.7	4.5
	5% WB	MCDBR	4.7	6.2	8.3	10.8	11.0	11.5	11.7	11.0	10.0	8.1	5.4	5.1
		MCWBR	4.3	5.2	6.1	6.2	6.6	6.2	6.4	6.0	6.4	5.8	4.7	4.7
Clear-Sky Solar Irradiance	taub	0.319	0.344	0.381	0.413	0.411	0.408	0.428	0.422	0.385	0.367	0.343	0.313	
	taud	2.327	2.310	2.234	2.211	2.265	2.294	2.251	2.281	2.361	2.381	2.374	2.331	
	Ebn at Noon	655	752	794	816	840	845	818	800	786	717	622	589	
	Edh at Noon	61	83	109	127	126	124	127	117	96	77	59	51	
All-Sky Solar Radiation	RadAvg	0.62	1.30	2.39	3.99	4.96	5.25	4.91	4.19	3.06	1.68	0.75	0.45	
	RadStd	0.07	0.21	0.27	0.54	0.59	0.41	0.67	0.34	0.34	0.21	0.09	0.04	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (43 neighbors)		+0.38	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page